EDUPATH BD

Production Release Candidate

Feature, technology and manual acceptance guide

Release field	Value
Version	1.0.0-rc.2 / Android+iOS build 21
Prepared	16 September 2026
Purpose	Navigate the full app, compare expected behavior, capture defects and prepare real production evidence.
Scope	Teacher requirements 1-24, professional additions, old-release merges, tools, security, recovery and launch gates.

This manual separates three things clearly: code that is implemented and tested, behavior that must be verified on the final laptop/phone, and external production evidence that cannot be proven without live accounts, infrastructure or Windows/macOS tooling.

Use it as a pass/fail workbook. Do not mark a feature complete because the screen exists; confirm authorization, saved state, reload behavior, errors and the expected API result.

1. Release verdict and evidence

Current baseline	v1.0.0-rc.2 merges useful v0.9.2, v0.9.4 and v0.9.5 production features into the stronger v1 RC base.	Source complete; external deployment evidence remains.
Automated QA	TypeScript, 20 API/unit tests, 4 database-operations tests, TensorFlow verification, full API contract, release/production gates and build pass.	Green in the review workspace.
API acceptance	Profile persistence/export, account deletion, eligibility, scholarships, workflow, community, media, AI, admin, moderation and reporting are exercised.	All verifier checks pass.
Browser E2E	Desktop/mobile suite now opens responsive auth and Account & privacy controls.	Run on laptop: review workspace could not obtain Chromium binary.
Production claim	Strict startup, readiness, migrations, least privilege, encrypted backup/restore and integration probes are implemented.	Still a release candidate until external gates have dated evidence.

Recommended manual order

- Health and responsive shell
- Public catalog and eligibility
- Student identity and saved workflow
- Community/media in two sessions
- Admin publishing and reports
- Offline/native behavior
- External production and recovery proof

Always capture route, role, exact steps, expected result, actual result, screenshot, console error and failed network request for every defect.

2. Roles and architecture

Guest	Public catalog, pathways, verified circulars/media, maps, PWA shell and diagnostics.	Private student and all admin data.
Student	Profile, eligibility, prediction, shortlist, comparison, applications, reminders, scholarships, comments, votes, ratings, reports, AI, export and self-deletion.	Admin content, moderation and analytics.
Moderator	Moderation queue and allowed review actions.	Owner/admin operations outside assigned policy.
Admin	Drafts, verification, media, CSV import, analytics, reporting, moderation and audited changes.	Validation/source/audit bypass and self-service account deletion.

System layers

React + TypeScript + Vite	Responsive UI, forms, localization, lazy feature modules and build.
Tailwind + Framer Motion	Design system, accessible responsive layout and transitions.
Capacitor + PWA	Android/iOS packaging, local notifications, safe areas, installability and offline shell.
Express 5 + Zod	REST routing, request validation, authorization, rate limits and business workflows.
Apollo GraphQL	Modern filtered institution query alongside REST endpoints.
PostgreSQL + Prisma	Users, profiles, catalog, rules, applications, discussions, media, moderation and audit data.
Socket.IO + SSE	Live discussion, vote and rating updates.
TensorFlow.js + Gemini	On-device fit model plus source-aware generative guidance/fallback.
Docker + GitHub Actions + Playwright	Repeatable migrations, hardened runtime, CI QA and browser acceptance.

3. Product shell, accessibility and offline behavior

Responsive navigation	Desktop header, mobile menu, bottom navigation and footer reach every major section with no duplicate IDs or overflow.	Check 360, 411, 768 and desktop widths; open/close menu; follow all links; browser Back.	Hidden auth/menu, clipped content, wrong active state or dead anchor.
Bangla/English	Visible labels and persistent locale change across reload and restart.	Switch twice; reload; open auth, workspace, diagnostics and errors.	Mixed labels, lost preference or untranslated validation.
Accessibility	Skip link, single visible H1, named controls, focus, reduced motion and 40-44 px touch targets.	Keyboard-only pass; inspect control names; enable reduced motion; run QA panel.	Focus trap, icon without name, tiny target or motion ignoring preference.
PWA/offline	Installable shell and cached public catalog after an online visit; private responses are never cached.	Visit online, disconnect, reload, inspect caches and signed-in state.	Blank shell, stale private data or false live status.
Diagnostics	Health, API, navigation, storage, viewport and captured runtime/network errors can be exported.	Run quality audit and Device check on laptop and phone; stop API and rerun.	False green state, localhost in APK or secrets displayed.

4. Identity, sessions, profile and privacy

Username reservation	Debounced availability and registration use the same store; unique collision is rejected and failed Firebase registration rolls back.	Try invalid/existing names and two-session collision.	Name says available then is silently stolen, or orphan identity remains.
Email/phone verification	Firebase verified email or phone creates an EduPath session; phone fictional-number mode is demo-only.	Test email link, fictional number and real project separately.	Unverified identity receives protected access or test bypass leaks to production.
JWT and roles	Short-lived issuer/audience checked app JWT with server-side student/moderator/admin guards.	Tamper token, wait expiry, call admin route as student and reload.	UI-only protection, infinite expired-session loop or protected response.
Cloud profile sync	Authenticated pathway profile hydrates after login and saves with loading/saving/synced/error state.	Change GPA/group/location, wait for Synced, reload and sign in on second browser.	Local value falsely labelled synced or older cloud state overwrites new input.
Personal export	Only the signed-in user's profile, shortlist, applications and activity download as JSON; no Firebase UID/password/secret.	Create state, export, inspect file and try another account.	Cross-account data or credential material in export.
Account deletion	Student types username; production also requires matching Firebase identity authenticated in last 10 minutes. Device state/reminders are cleared after success.	Wrong confirmation; old sign-in; fresh sign-in; reload and check username availability.	Admin deletion exposed, partial identity deletion or old session remains usable.
Sign out	Firebase session, app JWT and protected UI are cleared.	Sign out desktop/mobile; reload; revisit protected controls.	Workspace remains mounted or previous token still works.

5. Discovery, eligibility and intelligence

Institution/program catalog	Verified records only; search, type/group/district filters and server pagination combine.	Use each filter alone/together; page forward/back; open official source.	Draft leak, count mismatch, filter reset or broken source.
Circular deadlines	Verified sourced circulars follow date range, fee and ordering rules.	Test valid/invalid ranges, boundaries and empty result.	Timezone day error, inverted range accepted or source missing.
Eligibility engine	Server evaluates current database admission rules by level, group and GPA and returns reasons plus sourced programs.	Try eligible/ineligible Science, Humanities, Business and SSC technical profiles.	Client-only formula, stale result or eligibility confused with prediction.
TensorFlow fit	Bundled trained 5-8-1 model runs on-device/offline and ranks a strong compatible profile higher.	Load once, disconnect, compare strong/weak inputs and repeat.	Network dependency, random result or admission guarantee claim.
Why recommended	Recommendation shows profile factors, reasons and disclaimer.	Compare multiple pathways with different inputs.	Only a score, no reasons, or certainty language.
Edu AI and summaries	Authenticated source-aware answer cites records; Gemini, demo and safe fallback modes are labelled. Circular summary stays tied to its source.	Ask a sourced question with/without key; inspect mode and citations.	Uncited claim, hidden fallback, secret leak or summary of wrong circular.

6. Student planning and everyday workflow

Three-way comparison	Up to three institutions/programs persist locally and display exact comparable fields.	Add/remove/reorder three; attempt fourth; reload.	Duplicates, fourth item, stale or mismatched fields.
Shortlist	Signed-in shortlist persists server-side and is isolated per account.	Save/remove, reload, then inspect second student.	Guest mutation or cross-account result.
Application tracker	Program application creates valid status workflow and document checklist.	Create, advance status, toggle documents, reload.	Invalid jump, duplicate application or lost checklist.
Deadline reminders	Browser/native reminder can schedule and cancel; denied permission degrades safely.	Allow/deny, schedule near reminder, cancel and restart phone.	False success, duplicate alert or crash on denial.
Scholarship matching	Level/group/GPA query returns verified sourced opportunities and explains match.	Test high/low GPA and SSC Vocational/HSC groups.	Unverified result, missing source or same list for all profiles.
Parent progress PDF	Student progress report downloads and shares using native sharing when available.	Create workflow state, export PDF, inspect totals and share/cancel.	Wrong student, missing state, unreadable PDF or share crash.
Career pathway	Career outcome text and pathway cards connect interests, study options and fit reasons.	Change interest, open pathways and verify result changes.	Generic unchanged outcome or misleading job guarantee.

7. Community, media and map

Comments/replies	Authenticated database writes appear live; nested replies retain parent and author.	Use two sessions; post comment/reply; reload both.	Optimistic UI without persistence, wrong parent or private data leak.
Like/dislike	One vote per user/comment, server totals and live update; changing vote reconciles totals.	Vote from two users, switch type, reload.	Double count, client-only count or user can vote repeatedly.
Institution rating	One 1-5 rating per account with live average, count and distribution.	Rate from two sessions, edit score, reload.	Average drift, score outside range or self-duplication.
Reports/moderation	Student reports enter protected moderation queue; decision and resolution are audited.	Report comment, open admin queue, resolve, refresh student/admin.	Student sees queue, report disappears or decision has no audit.
Campus media	Verified institution media wins; otherwise language-aware global fallback, bundled admission video and device speech fallback.	Select several campuses; toggle language; block network audio; inspect fallback.	Old neural-network placeholder, empty panel or unverified media shown.
Map/location	Leaflet/OpenStreetMap works without key; optional Google embed; location finds nearest listed campus.	Allow/deny geolocation; choose campus; test map controls on phone.	Permission crash, tiny controls, wrong campus or mandatory paid key.
Upload validation	PDF/image/audio requires auth, size/MIME/base64/signature validation and safe preview/storage.	Upload valid, spoofed, oversized and unsupported files.	Generic error, executable accepted or unverified upload shown publicly.

8. Admin, reporting and content lifecycle

Admin bootstrap	One-time production command accepts only verified Firebase email, explicit confirmation, UID-safe linkage and audit log.	Run without flag/demo mode/wrong email, then valid production staging run.	Default admin password, role change without audit or conflicting UID overwrite.
Draft and verification	Institution/program/circular/media drafts remain hidden until reviewed; child visibility follows parent state.	Create draft, verify child first, verify parent, mark outdated.	Draft leak, orphan verified child or missing source/reviewer.
Bulk CSV import	Protected import parses quotes, reports row errors and saves valid rows as drafts.	Import valid, quoted commas, duplicate slug and bad enum/date rows.	Whole file silently fails or invalid rows publish.
Analytics	Admin metrics aggregate applications, ratings, comments, reports and AI activity.	Generate activity, refresh dashboard and compare counts.	Client-estimated numbers, stale totals or student access.
Reporting	Protected JSON/CSV plus XSD, PostgreSQL views and C# Crystal viewer scaffold support the final Windows report.	Download data; open viewer assets; validate schema/sample.	Public export, secret column or claim that missing .rpt is complete.
Audit trail	Admin content edits, publication and moderation decisions record actor/action/entity and before/after data.	Perform changes and inspect audit records in production DB/staging.	Unattributed mutation or secrets in audit payload.

9. Teacher requirements 1-24 traceability

1	Scalable PDF ERD	Vector A3 ERD and relationship matrix	Source complete
2	Real-time own-DB comments	PostgreSQL/memory comments, replies, SSE/Socket	Live DB two-client proof
3	Live like/dislike	Unique per-user vote and server totals	Live DB two-client proof
4	Username check	Debounce, reservation, unique constraint, rollback	Live Firebase/DB proof
5	Email and phone verification	Firebase email and genuine phone OTP plus fictional demo	Firebase project proof
6	Inputs/dropdowns	Student and protected admin forms	Complete
7	Real-time rating	Per-account rating, live average/distribution	Live DB proof
8	Pagination	Server pages and mobile controls	Complete
9	External audio/video	Verified YouTube/audio and safe fallbacks	Complete
10	Maps/location	Leaflet/OSM, geolocation, optional Google	Device proof
11	Session/JWT	Firebase token then role-guarded app JWT	Complete
12	On-device prediction	Committed TensorFlow.js 5-8-1 model	Model verified
13	Cloud upload/display	Signature validation and Supabase adapter	Live Supabase proof
14	Datepicker	Shared React Datepicker and date-order rules	Complete
15	Animation	Framer Motion transitions	Complete
16	Styling	Tailwind responsive system	Complete
17	Git/Vite	Workspaces, Vite and one-command QA	Hosted Git evidence
18	GraphQL	Apollo institution query	Verified
19	REST	Express health/auth/catalog/workflow/admin APIs	Verified
20	Crystal Report	JSON/CSV/XSD/views/C# viewer	Final Windows .rpt pending
21	Modern frontend	React/TS/Capacitor Android+iOS+PWA	iOS signing pending
22	Docker/CI/CD	Migrations, hardened runtime, Compose, Actions	Hosted/runtime evidence
23	Generative AI	Gemini adapter, citations, audit, fallback	Live Gemini key proof
24	Additional features	Localization, offline, accessibility, workflows, scholarship, PDF, CSV, admin	Implemented; manual acceptance

10. Tools and technologies - what each one builds

React 19	Component UI	Shell, modals, dashboards, forms and feature hubs.
TypeScript	Static safety	Shared contracts, safer refactors and CI type gate.
Vite/Rolldown	Web build	Fast development, production assets, proxies and explicit chunk groups.
Tailwind CSS	Design system	Responsive spacing, color, typography and touch targets.
Framer Motion	Interaction	Entrance/results/modal transitions with reduced-motion compatibility.
Capacitor	Native bridge	Android/iOS wrapper, safe areas and native plugin access.
Local Notifications	Native reminders	Deadline scheduling/cancellation on supported devices.
Service worker/PWA	Offline delivery	Installable shell and safe public API cache.
Express 5	Backend API	REST, middleware, errors, health/readiness and graceful shutdown.
Zod	Validation	Bounded request schemas and field-specific errors.
Apollo/GraphQL	Modern query API	Filtered institution query.
PostgreSQL	System of record	Accounts, catalog, workflow, community, audit and analytics.
Prisma	Data access/migrations	Typed queries, relationships, migration and seed workflows.
Firebase Auth/Admin	Identity	Email/phone verification, UID linkage and identity deletion.
Supabase Storage	Cloud files	Validated media storage and display adapter.
TensorFlow.js	Local ML	Offline pathway-fit inference with committed weights.
Gemini	Generative AI	Source-aware guidance and circular summary; fallback when unavailable.
Leaflet/OSM	Maps	Key-free campus map and device-location comparison.
Socket.IO/SSE	Realtime	Live community, vote and rating updates.
jsPDF/html2canvas	Student reports	Parent progress PDF generation and sharing.
React Datepicker	Dates	Deadline range and admin admission date inputs.
Docker/Compose	Deployment	PostgreSQL, migration, API and isolated recovery jobs.
GitHub Actions	CI	Generation, QA, E2E and critical audit gates.
Playwright	Browser acceptance	Desktop/mobile navigation, API and student-account smoke flows.
Crystal Reports/C#	Formal reporting	Windows viewer scaffold for the final proprietary .rpt layout.

11. Production runtime, recovery and security

Strict production startup	Fails closed without PostgreSQL, strong JWT secret, HTTPS origins, Firebase and Supabase credentials.
Health/readiness	Liveness proves process; readiness proves data store; response includes request ID for tracing.
Migrations	Separate deployment job runs Prisma migrations before API; production startup never seeds.
Least privilege	Unprivileged, read-only container with dropped capabilities; database has no public port.
Encrypted backup	pg_dump custom archive encrypted with AES-256/PBKDF2, SHA-256 manifest and bounded retention.
Safe restore	Dry run by default; checksum/archive validation; exact DB confirmation plus explicit production approval.
Integrity checks	Ranges, dates, sources, duplicate identities and verified media rules; strict mode promotes warnings.
Live integration check	Production-only PostgreSQL, Firebase Admin and Supabase bucket probes without printing secrets.
Runtime check	HTTPS liveness/readiness/version/CORS/catalog validation against deployed URL.

Core commands

```
npm run qa
npm run test:e2e
npm run audit:critical
npm run production:integrations-check
npm run production:runtime-check -- --api-url https://api.example.com --origin https://app.example.com
npm run db:backup
npm run db:restore -- --file FILE.dump.enc
npm run db:integrity -- --strict
```

Never test a production restore on the only database. Restore into a separate drill database, run strict integrity checks and record recovery time and data loss.

12. Remaining external evidence before general availability

PostgreSQL	Migrate non-demo DB and run full integration workflow	Deployment owner + dated QA log
Catalog	Load current official institutions/programs/circulars/scholarships with sources and review dates	Content owner approval
Firebase	Authorized domains, email/phone flows and recent-auth deletion	Physical-phone/video evidence
Supabase	Private bucket policy plus upload/read tests	Integration check and manual upload
Gemini	Live key, quota, citations and fallback	Response evidence without key exposure
Domain/TLS	Public HTTPS frontend/API and reverse proxy	Runtime checker output
Android	Release keystore backup, signed AAB and physical-device sheet	Store-ready artifact
iOS	Xcode signing/archive and physical-device pass	macOS/Xcode evidence
CI	Hosted final commit with green QA/E2E/audit	GitHub Actions link/screenshot
Recovery	Scheduled off-site encrypted backup and separate-DB restore drill	Manifest and drill report
Monitoring	Availability, latency and error alerts	Hosting dashboard evidence
Privacy/legal	Privacy policy, terms, retention/export/deletion approval	Product-owner sign-off
Crystal Reports	Design and open final .rpt in SAP extension on Windows	Screenshot and included file

Until every row has an owner, date and evidence, call the project a production release candidate - not a generally available public service.

13. Final manual acceptance run

1	Fresh install/open	No blank screen; health and version rc.2 are correct
2	Responsive + language	No overflow; all nav/auth/account controls; locale persists
3	Student identity	Registration/login/verification/session expiry/signout correct
4	Profile + privacy	Cloud sync, export, cache clear and recent-auth deletion correct
5	Catalog	Search/filter/pagination/source links correct
6	Eligibility + ML	Rules explain eligibility; offline model is deterministic
7	Compare/shortlist	Limits, persistence and account isolation correct
8	Applications	Statuses/checklist/reminders survive reload
9	Scholarship + parent PDF	Matches are sourced; PDF/share content is accurate
10	Community	Two-session comments/replies/votes/ratings update live
11	Media/map/upload	Verified fallback precedence, location denial and upload errors correct
12	AI	Mode and citations are visible; failure falls back safely
13	Admin	Draft/publish/outdated hierarchy, CSV errors and audit behavior correct
14	Reports	Analytics, moderation, JSON/CSV and Crystal inputs correct
15	Offline/native	Public shell/cache, notifications, safe area, Back and rotation correct
16	Production ops	Integration/runtime checks, backup, dry-run restore and integrity evidence captured

Laptop commands

```
npm ci
npx playwright install chromium
npm run qa
npm run test:e2e
npm run audit:critical
```

Phone build (replace the IP with the current Wi-Fi address)

```
CAPACITOR_ALLOW_MIXED_CONTENT=true npm run android:debug -- --api-url http://192.168.x.x:4000 --install
```

14. Defect capture and old-version merge record

Defect record template

ID/severity	Example: RC2-MOB-001 / Blocker, High, Medium, Low
Environment	Version, browser/device, viewport, network, API mode
Role/state	Guest/student/admin; fresh/reloaded/offline
Steps	Small numbered reproduction with exact input
Expected vs actual	One measurable sentence each
Evidence	Screenshot/video, Console text, request URL/status, QA JSON
Regression	Whether v0.9.2/v0.9.4/v0.9.5 behaved differently

Selective merge result

v0.9.2	Production hardening concepts	Current v1 strict configuration, readiness, containers and code splitting were stronger and retained.
v0.9.4	Firebase UID, profile/privacy and account lifecycle	Merged, then strengthened with recent Firebase authentication and identity-match deletion checks.
v0.9.5	Admin bootstrap and live integration/runtime probes	Merged and adapted to current health/readiness/version contracts.
Current v1 recovery	Encrypted backup, restore dry run, retention, integrity and recovery docs	Retained; old runtime files were not allowed to weaken it.

Release handoff

- Keep the rc.2 source archive, checksum, release notes and this manual together.
- Run E2E on the laptop after installing Playwright Chromium and attach the HTML report.
- Use the in-app QA JSON plus this manual to report every manual discrepancy.
- Do not delete the final source archive until it has been copied to two separate storage locations.

End of manual